ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Abstract

Can a compact 3B vision-language model solve multi-step reasoning questions over complex scientific charts? We investigate this question by benchmarking `Qwen2.5-VL-3B-Instruct` on 500 challenging reasoning problems from the CharXiv benchmark. Rather than relying solely on outcome accuracy, we apply dense process-level verification via a multimodal LLM-as-a-judge (`muse-spark-1.1`) to trace where reasoning trajectories diverge. We find that rollouts fail early: over 80% of initial mistakes occur within the first two steps, and an error at step N leads to downstream failure in $\sim$83% of cases. Text analysis of 2,920 judge critiques reveals that failures are overwhelmingly perceptual (axis and legend misreadings comprise 43.5% of errors) rather than logical or arithmetic (1.3%). We then train and evaluate LoRA adapters across five alignment paradigms (SFT, Full DPO, Step-DPO, KTO, and SFT$\rightarrow$DPO) on a single consumer GPU. Full-sequence DPO achieves the highest exact-match accuracy at 29% (improving over the 26% base), while SFT enforces 100% formatting compliance at the cost of reasoning accuracy (23%). Although KTO captures the ground-truth answer in text 66% of the time, it suffers from structural formatting collapse. Our findings demonstrate that for small VLMs, the primary bottleneck in chart reasoning is visual grounding rather than chain-of-thought logic.

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